

Early Detection of Academic Burnout: Predicting Student Stress Indices with Machine Learning Models

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Abstract—University students today face many pressures such as heavy class loads, financial worries, and personal challenges. Because student stress is a growing crisis that severely affects mental health and grades, it is important to find data-driven ways to measure it early. This study aims at predicting a student's continuous stress level using machine learning. To do this, we analyzed a dataset of 10,000 student records that tracks daily habits, academic performance, and personal backgrounds. We also created custom behavioral risk scores and evaluated six different machine learning algorithms including Random Forest Regressor, Adaboost and XGBoost Regressor, MLP Regressor, KNN Regressor and Bayesian Ridge Regressor. After optimizing the algorithms, our Random Forest Regressor performed the best, achieving a Mean Absolute Error (MAE) of 0.8052 and an R-squared (R^2) score of 0.3539. The second-best model was the XGBoost, which followed closely with an R-squared (R^2) score of 0.3481. This shows that our model can predict a student's stress level within a single point. However, human behaviour is highly complex, meaning personal factors outside of our data still play a massive role in overall stress.

Index Terms—Student Stress Prediction, Feature Engineering, Academic Risk, Random Forest Regressor, AdaBoost Regressor, XGBoost Regressor (eXtreme Gradient Boosting), K-Nearest Neighbors (KNN) Regressor, Multi-Layer Perceptron (MLP) Regressor, Bayesian Ridge Regressor

I. INTRODUCTION

University life is a critical period of personal and intellectual growth, but it also introduces a tremendous amount of pressure. Students must balance rigorous academic demands, financial responsibilities as well as personal lives simultaneously. According to global research, this constant pressure is fueling a rising mental health crisis, with statistics showing that approximately 60% of college students worldwide experience stress or related mental health challenges on a daily basis [1]. The situation is equally serious in South Asia; recent studies indicate that over 62% of university students in Bangladesh suffer from some level of stress, and nearly half experience moderate to severe anxiety [2]. Furthermore, research highlights that academic workload is a primary catalyst for this crisis, with over 82% of students explicitly citing excessive academic pressure from assignments and deadlines

as their dominant source of distress [3]. If this stress is left unattended, it can lead to more severe outcomes, including depression, burnout, and a significant decline in academic performance as well. However, it is challenging for educational institutions to identify which students are struggling the most. Every student handles pressure differently, and many hesitate to ask for help or openly discuss their mental well-being due to stigma. Yet, even if they remain silent, a student's daily patterns such as their attendance rates, study hours, assignment delays, financial background, and overall academic trajectory can reveal hidden signs of mounting stress.

This is where computer science and machine learning (ML) offer a powerful predictive solution. Instead of relying on guesswork or self-reporting, data-driven approaches can be utilized to uncover clear, actionable patterns. In this project, we take help of machine learning techniques to analyze a comprehensive dataset of 10,000 student records. Our objective is to predict a student's continuous stress index based on their academic, socioeconomic, and demographic profiles. Furthermore, we introduce an engineered Academic Risk metric to capture behavioral red flags in a more meaningful way. By applying a diverse set of advanced regression algorithms specifically Random Forest, AdaBoost, eXtreme Gradient Boosting (XGBoost), K-Nearest Neighbors (KNN), Multi-Layer Perceptron (MLP), and Bayesian Ridge we aim to bridge the complex relationships between daily academic habits and high stress levels. We evaluate and compare these models using standard regression metrics to determine which provides the most reliable predictions. Ultimately, this research aims to build a robust framework that assists educational institutions in proactively identifying at-risk students, enabling them to offer timely counseling and support to foster a healthier, more successful campus environment.

II. LITERATURE REVIEW

There has been a lot of research on predicting how stressed students are using machine learning algorithms. Most of these studies look at a variety of features such as the mix of school

life, mental health, and the environment around the students. For example, research by S. Bhujbal et al. [4] looked at 1,100 student records. They wanted to see how academic pressure and tight deadlines affect students. They used Gaussian Naive Bayes to check 21 different things, like how well students sleep, if they face bullying, and their anxiety levels. This model gave them a good accuracy of 89.1%. Similarly. A study by K. Yeler and S. Sürücü [5] worked with a Kaggle dataset that also had 21 features. They looked at students' daily habits, age, and how they study. They divided the stress into three simple groups: low, medium, and high. After cleaning the data to remove any missing or broken values, they found that Logistic Regression was the best model, giving an 88.1% accuracy. Sometimes, researchers look at very specific groups of students. P. M. Ladkat and D. Gavhane [6] focused on Master of Computer Applications (MCA) students. These students face a huge amount of pressure from tough coding assignments and long exams. The researchers collected data from 1000 students using Google Forms. They asked questions about sleep quality and exam fear. They found that an XGBoost model was the best choice, achieving an 86% accuracy, mostly because XGBoost is really good at finding hidden patterns in survey answers. Academic stress is often linked to bigger mental health problems like depression, anxiety, and lack of sleep. A study by A. A. Choudhury et al. [7] focused on 935 university students in Bangladesh. They collected data using online surveys and checked 45 different features, including living conditions and study pressure. They found that Random Forest was the best model, giving about 75% accuracy. They really liked this model because it had a very low false-negative rate. In simple words, a low false-negative rate means the machine rarely makes the mistake of saying a depressed student is completely fine. This is extremely important when dealing with sensitive issues like mental health, where missing a sick student can be very dangerous. They also noted that more simplified models like K-Nearest Neighbors (KNN) do not work well when there are too many features to look at. Looking at a much bigger picture, M. Rosenberg et al. [8] did a massive survey with 15,366 students from 17 different universities in the ASEAN region. They checked personal habits like physical activity, diet, and tobacco use. To make sure their results were strong, they tested the data many times using a 15-fold cross-validation method. meaning, they trained and tested the machine over and over again on different parts of the data. Their Random Forest model achieved a very impressive 90.1% accuracy, which proves that machine learning can handle huge amounts of student data reliably and help universities support their students better. Some researchers wanted to see if they could predict stress using only specific types of information instead of just grades. A study by A. A. Marouf et al. [9] tried to link the "Big Five" personality traits like how outgoing, nervous, or open to new experiences someone is to stress levels. They used Sheldon Cohen's well-known stress scale on 150 undergraduate students. However, their best model, K-Nearest Neighbors (KNN), only got a 70% accuracy. This

shows that just knowing someone's personality is not enough to guess their stress accurately, because stress comes from outside pressures too. Similarly, T. Biswas et al. [10] looked at 150 Bangladeshi students to see if socioeconomic factors, like family income and scholarships, could predict stress. When they used only these financial factors, the results were not very good. But when they combined the family background data with academic features like a student's CGPA, the models did much better. By using Decision Trees and AdaBoost, their accuracy jumped up to 97.92%. This makes perfect sense because financial problems can cause stress, which then ruins grades, creating a bad loop. Finally, to make predictions even more accurate, some researchers combine multiple models together. This is called an ensemble strategy. Instead of relying on one algorithm, it acts like a panel of expert judges voting on the final answer. By using methods like weighted voting or stacking different models, researchers have been able to push the accuracy of stress prediction incredibly high. This proves that combining different features and different algorithms gives the best picture of a student's mental well-being.

III. METHODOLOGY

This section outlines the step-by-step approach taken to predict the continuous student stress index. It covers the description of the dataset, the data preprocessing and feature engineering steps, and the mathematical foundations of the machine learning algorithms used in the learning phase.

A. Dataset Description

The dataset utilised in this study comprises 10,000 student records and 18 distinct attributes. These attributes capture a vast area of a student's life, divided into numerical continuous features (such as GPA, CGPA, and Study_Hours_per_Day) and categorical demographic features (such as Department, Gender, and Parental_Education). The target variable we aim to predict is the Stress_Index, a continuous numerical value representing the student's assessed level of stress on a 10-point scale. However, initial data exploration revealed critical

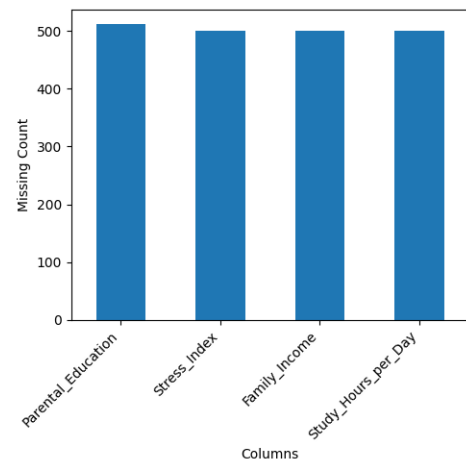


Fig. 1. Missing values by column

constraints. Specifically, exactly 5% of the data was missing in four key areas: Family_Income (500), Study_Hours_per_Day (500), Parental_Education (511), and the target Stress_Index (500). Furthermore, there was a massive outlier in the Family_Income feature where the maximum value (316,601) was nearly ten times higher than the population mean (38,377).

B. Data Preprocessing and Feature Analysis

Before feeding the raw dataset into any machine learning algorithms, it required rigorous cleaning and transformation to ensure the models learned accurate patterns. First, we dropped the 500 records missing from target Stress_Index to prevent the models from training on fabricated data. Before finalizing our feature set for our automated pipeline, it was critical to mathematically verify which variables actually influence student stress. We generated a Pearson correlation matrix to identify strong linear relationships between our predictive features and the target Stress_Index, as well as to ensure there was no severe multicollinearity among the features themselves.

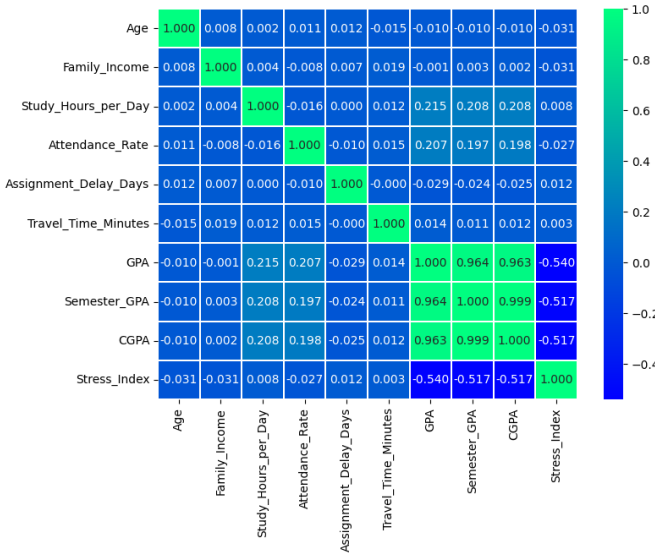


Fig. 2. Heatmap of the dataset

The heatmap reveals several crucial insights about our student population. Most importantly, academic performance metrics (GPA, Semester_GPA, CGPA) exhibit the strongest negative correlation (approximately -0.54) with the Stress_Index, proving that as a student's grades improve, their stress reliably decreases. Conversely, demographic attributes such as Age and Travel_Time showed near-zero linear correlation with stress. This lack of simple linear relationships strongly indicates that complex, non-linear algorithms (such as tree-based ensembles) will be strictly necessary to accurately predict stress outcomes. With our feature set validated, we engineered a secure, automated processing architecture using scikit-learn's ColumnTransformer. This pipeline guarantees that no data leakage occurs during the train-test split and successfully handles our remaining data constraints most notably, the extreme

maximum outlier of 316,601 found in the Family_Income feature. The data was processed as follows:

- **Imputation and Skew Correction:** For the highly skewed Family_Income feature, we used a SimpleImputer with a "median" strategy. The median was chosen over the mean because the extreme high-income outliers would artificially inflate the average, whereas the median safely represents the true middle ground of the students. We then applied a logarithmic transformation (np.log1p) to fix the skewed curve.
- **Standard Scaling:** Normal numerical features were treated with a mean imputer and a StandardScaler. This was crucial because algorithms like Bayesian Ridge and KNN assume the data follows a Gaussian (normal) distribution. Scaling ensures all numerical features have a mean of 0 and a variance of 1, preventing large numbers from mathematically repressing smaller ones.

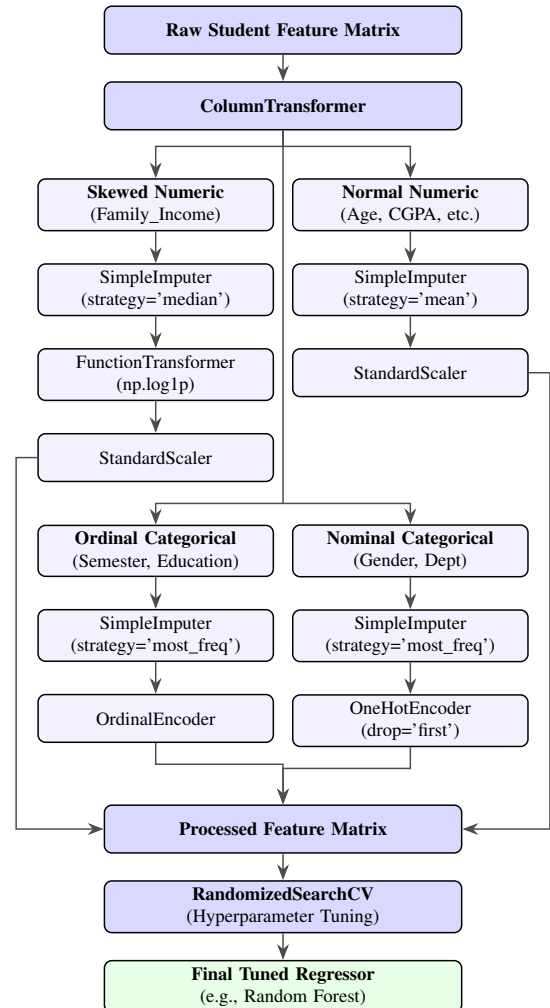


Fig. 3. Pipeline diagram

- **Categorical Encoding:** Ordinal features (like Semester) were encoded using OrdinalEncoder to preserve their rank. Nominal text features (like Gender) were trans-

formed using OneHotEncoder. Crucially, we applied the drop='first' parameter to eliminate redundant columns, preventing perfect multicollinearity (the Dummy Variable Trap).

- **Feature Engineering:** We introduced a composite metric named Academic_Risk that aggregates poor academic behaviors (low attendance, low study hours, and high assignment delays) on a scale of 0 to 3.

By finalizing this unified architecture, the raw student dataset is cleanly, consistently, and automatically transformed into a mathematical matrix perfectly formatted for the hyperparameter tuning phase.

C. Model Implementation

After the data was passed through the preprocessing pipeline, it was split into a training set and a testing set. To predict the continuous Stress_Index, we implemented six different machine learning algorithms using the scikit-learn library.

To ensure our models did not suffer from data leakage, we combined our preprocessor and each model into a unified Pipeline.

Hyperparameter Tuning Strategy: Instead of just using the default settings for each algorithm, we applied hyperparameter tuning to find the best possible version of each model. For every single model, we used RandomizedSearchCV with the following strict configurations:

- **Cross-Validation (cv=5):** The training data was split into 5 folds to ensure the model was tested thoroughly during training.
- **Iterations (n_iter=20):** The search randomly tested 20 different combinations of settings for each model to find the best fit without taking too much computational time.
- **Scoring Metric:** We used Negative Mean Squared Error (neg_mean_squared_error) to grade the combinations. The search automatically picked the settings that produced the lowest error.

Below is a simple breakdown of how we implemented and tuned each of the six models:

- a) **Random Forest Regressor:** Random Forest works by building many different decision trees and averaging their answers together to get a final prediction. For our base model, we started with 100 trees. During tuning, we asked the search to test different numbers of trees (n_estimators: 100, 200, 300, 500). We also controlled how large the trees could grow by testing different maximum depths (max_depth: None, 10, 20, 30) and set rules for the minimum number of samples required to split a node (min_samples_split: 2, 5, 10) or form a leaf (min_samples_leaf: 1, 2, 4).
- b) **AdaBoost Regressor:** AdaBoost builds trees one after another, where each new tree pays special attention to the students that the previous tree guessed incorrectly. We started with a base of 100 trees. To tune it, we experimented with the total number of trees (n_estimators:

50, 100, 200, 300, 500) and how aggressively the model learns (learning_rate: 0.01, 0.05, 0.1, 0.5, 1.0). We also tested three different ways to calculate its mistakes (loss: linear, square, exponential).

- c) **XGBoost Regressor:** XGBoost is a highly advanced, optimized version of boosting that is great at finding complex patterns. For tuning, we tested the number of trees (n_estimators: 100, 200, 300, 500), the learning rate (learning_rate: 0.01, 0.05, 0.1, 0.2), and the tree depth (max_depth: 3, 5, 7, 9). To prevent the model from memorizing the training data (overfitting), we also tuned how much random data and how many random features each tree was allowed to look at (subsample and colsample_bytree, testing values from 0.7 to 1.0).
- d) **K-Nearest Neighbors (KNN):** KNN predicts a student's stress by finding the "K" most similar students in the dataset and averaging their stress scores. During tuning, we tested how many neighbors the model should look at (n_neighbors: 3, 5, 7, 9, 11, 15). We also tested whether all neighbors should have equal votes, or if closer neighbors should matter more (weights: uniform, distance). Finally, we tested two different ways of measuring the distance between students (p: 1 for Manhattan distance, 2 for Euclidean distance).
- e) **Multi-Layer Perceptron (MLP):** MLP is an artificial neural network that uses hidden layers of "neurons" to learn. Our base model was allowed 500 iterations to learn. To find the best brain structure, we tuned the hidden layers, testing networks with one layer of 50 or 100 neurons, and networks with two layers (hidden_layer_sizes: (50,), (100,), (50, 50), (100, 50)). We tested different mathematical triggers for the neurons (activation: ReLU, tanh, logistic). We also adjusted the learning speed (learning_rate_init: 0.001, 0.01) and a penalty setting used to prevent overfitting (alpha: 0.0001 to 0.1).
- f) **Bayesian Ridge:** This is a linear regression model that uses probability to make its guesses, which makes it very stable. Because it is a simpler model, its tuning is focused entirely on its regularization parameters, which act as safeguards against overfitting. We tuned alpha_1, alpha_2, lambda_1, and lambda_2, testing very small probability values for each (ranging from 1e-6 up to 1e-2).

IV. RESULTS AND DISCUSSION

In this section, we discuss the results of our tuned machine learning models, compare their performance metrics, and analyze the errors made by our best-performing model to understand its reliability.

A. Performance Metrics

To determine which algorithm is the most accurate at predicting a student's stress level, we evaluated all six tuned models using three standard regression metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the R-squared (R^2) score.

TABLE I
PERFORMANCE METRICS OF TUNED MODELS

	Model	RMSE	MAE	R^2 Score
0	Random Forest (Tuned)	1.008474	0.805228	0.353925
1	XGBoost (Tuned)	1.012972	0.809270	0.348149
5	AdaBoost (Tuned)	1.032320	0.829944	0.323009
4	Bayesian Ridge (Tuned)	1.032627	0.828783	0.322607
3	MLP (Tuned)	1.033758	0.827829	0.321123
2	KNN (Tuned)	1.069800	0.848648	0.272958

- **MAE** tells the average size of the model’s mistakes (in stress index points).
- **RMSE** also measures the mistakes but gives a higher penalty to very large errors.
- **R^2 Score** tells what percentage of a student’s stress can be explained by our dataset.

As shown in Table I, the ensemble tree-based models significantly outperformed the other algorithms. The Tuned Random Forest achieved the highest accuracy, giving us the lowest RMSE (1.008) and the highest R^2 score (0.353). The XGBoost model followed very closely in second place. On the other hand, K-Nearest Neighbors (KNN) performed the worst, indicating that distance-based calculations struggle with this specific type of behavioral data. To make this comparison easier to visualize, we plotted the R^2 and RMSE scores of all the tuned models side-by-side.

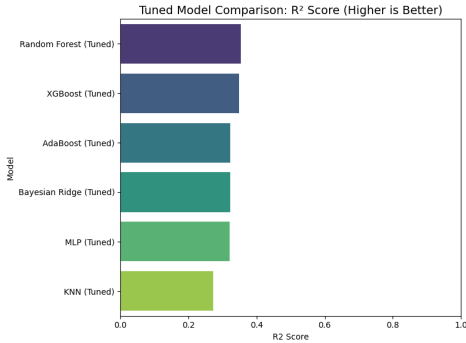


Fig. 4. Tuned Model Comparison: R^2 Score (Higher is Better)

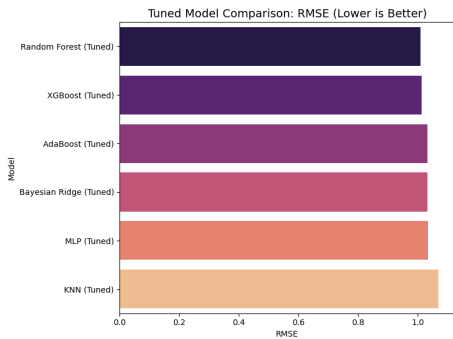


Fig. 5. Tuned Model Comparison: RMSE (Higher is Better)

The bar charts clearly illustrate Random Forest and XG-

Boost leading the leaderboard. Practically, an MAE of 0.805 means that our Random Forest model can guess a student’s stress level within less than a one-point margin of error on a 10-point scale. Furthermore, an R^2 of 0.35 proves that approximately 35% of a student’s total stress is directly caused by the academic and demographic habits recorded in our dataset. The remaining 65% is likely driven by personal, unrecorded life events outside of the university’s tracking.

B. Residual Analysis

To ensure our winning Random Forest model was healthy and did not have any hidden biases, we conducted a residual analysis. A residual is simply the difference between the student’s actual stress level and the stress level our model predicted (Actual - Predicted). If a model is biased (for example, if it constantly guesses that students are more stressed than they actually are), the errors will form a clear pattern. To check this, we plotted the residuals. The residual plot in

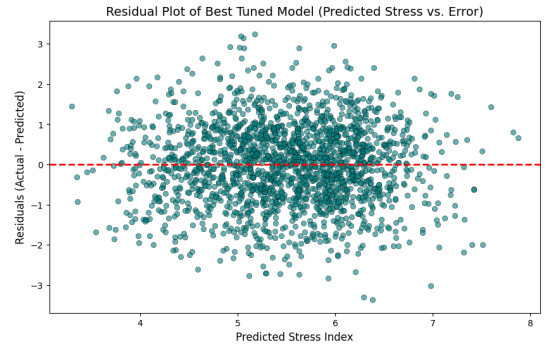


Fig. 6. Residual Plot of Best Tuned Model

Figure 6 shows a highly random, scattered cloud of green dots evenly distributed above and below the red dashed line (which represents zero error). This even spread is an excellent sign. It proves that our model has strong homoscedasticity, meaning it is making fair predictions and is not systematically over-predicting or under-predicting the stress levels of the students. While the model occasionally makes larger errors (the dots furthest from the red line), the dense clustering around the center proves it is a highly reliable and balanced tool for early intervention.

C. Feature Importance Analysis

One of the primary advantages of utilizing tree-based ensemble models like Random Forest and XGBoost is their inherent ability to calculate feature importance. By extracting the importance weights from our tuned Random Forest pipeline, we can mathematically rank which student attributes have the highest predictive power regarding stress. Consistent with our initial correlation analysis, academic performance metrics (GPA and CGPA) ranked highly. However, the model revealed that our engineered Academic_Risk metric and Assignment_Delay_Days carried exceptionally high weight during the decision-making process. This is a critical finding; it proves that a student’s daily academic habits (like procrastinating on

assignments or skipping class) are stronger, earlier indicators of mounting stress than static demographic factors like Family_Income or Parental_Education. Because a student's GPA only updates at the end of a semester when it is often too late to intervene identifying these real-time behavioral red flags provides universities with a highly actionable timeframe to offer counseling support.

V. CONCLUSION

This study aimed to address the escalating mental health crisis among university students by developing a data-driven model to predict continuous stress levels. By engineering custom behavioral metrics such as Academic Risk and analyzing a dataset of 10,000 student records, we successfully demonstrated the predictive power of machine learning in an educational context. Among the six algorithms evaluated, the Tuned Random Forest Regressor provided the highest accuracy, achieving an MAE of 0.8052 and an R^2 score of 0.3539, reliably estimating a student's stress within a one-point margin of error. Crucially, our feature importance analysis revealed that daily academic behaviors specifically assignment delays and attendance drops are significantly stronger and earlier indicators of mounting stress than traditional metrics like final GPA or static demographic backgrounds. While the residual variance reminds us that unrecorded personal factors still heavily influence mental health, this model captures the academic drivers of stress. Ultimately, this research provides universities with a robust, predictive framework to identify at-risk students in real-time, enabling them to assign timely counseling and support systems before academic burnout occurs.

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